

Edge Computing Enabled Smart Warehouse Management System for Food Processing Industries

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Abstract— Food safety is a primary concern across all global platforms, and food storage is a crucial and essential part of management. This clarifies that food warehousing should be fully automated or has some management system. Good food storage plays a vital role in food security, which is affected by food waste and food loss. As a result, it is critical to identify and remove microbial contamination before it spreads to other items. To end that, we propose an edge computing-enabled intelligent warehouse system with an ESP32 microcontroller and micropython. The food commodities studied in the other papers were fish, dairy, fresh food products, crop-field cucumber, seafood, wine, and pre-packaged food. The various gas sensors are used to detect microbial growth, food decay, and shelf life. Humidity and temperature sensors are employed to monitor the warehouse environment, along with RFID technology to track food items. Multiple nodes accessible via Telegram and dashboard notifications will be put in the building to enable law enforcement collaboration. These nodes will exchange information on the building's structure. Compared to the traditional warehouse management system, the proposed edge computing-enabled IoT warehouse shelves reduce waste and contamination in fruit and vegetable chambers. Designs are experimented with different sets of fruits and vegetables. Continuous live monitoring and an alert system demonstrate the effectiveness of the proposed innovative framework compared with an existing management system.

Keywords— Edge Computing, RFID, WSN, ESP32, IFTTT, IoT

I. INTRODUCTION

Several things can go wrong with a structure, and they all have the potential to impair its utility. Warehouses are infamous for having a high prevalence of common problems. Maintaining sufficient stock control is critical to avoid shortages, overstocks, missing items, and thievery. Some of the concerns stated above may affect customer satisfaction, raise labour expenses, and cause order processing to be delayed. When demand is overestimated or misinterpreted, there is a greater chance that a product will run out of supply or be overstocked. Order fulfilment

may be delayed, stock costs may rise, and revenue possibilities may be lost. When a facility is disorganised, it might be challenging to relocate objects, resulting in lengthier travels and traffic bottlenecks in high-traffic areas. This trend could lead to a decrease in output and an increase in labour costs. Errors can arise when technology is unused, staff expenses might rise, and wait times can lengthen. Automation may make warehouse tasks like picking, packaging, and inventory management more accessible and productive. When people do activities that endanger themselves or others, safety concerns arise. Security concerns can also lead to inventory and equipment damage. Inadequately trained warehouse staff may make costly mistakes and endanger themselves. Well-designed Training programmes can boost employee output and the number of errors committed. Outdated technology may limit the warehouse's capacity to respond to swings in client demand quickly and effectively. If technology is used effectively, it can help warehouse efficiency, accuracy, and adaptability. If these problems were addressed, the warehouse's efficiency, effectiveness, and customer happiness may all improve [1] [2].

Agriculture warehouses store and distribute agricultural products such as food, animals, and farming equipment. It is vital to store these objects in a specific manner in order to preserve their quality and value. Controlled storage conditions are commonly used in agricultural warehouses to prevent food spoilage, fungus growth, and insect infestation. This typically entails maintaining a specified temperature, relative humidity, and air quality range. This is critical for crops that are easily harmed by weather changes. Agricultural storage facilities may also provide value-added services such as sorting, grading, and packaging. Suppose a large number of people follow this technique. In that case, agricultural supply chains may become more efficient, making it easier for farmers and agribusinesses to meet the strict quality demands of their clients. IoT is based on the idea that a network of interconnected intelligent devices connected to the Internet

can efficiently share and receive data from the outside world [3] [4].

II. SMART WAREHOUSE MANAGEMENT SYSTEM

A modern and sophisticated warehouse, sometimes known as a "smart warehouse," enhances and streamlines warehouse operations by utilising cutting-edge technology such as the Internet of Things (IoT), automation, artificial intelligence (AI), and data analytics. Intelligent warehouses can improve supply chain management's efficiency, speed, and response to client needs. Sensors and other IoT devices are routinely used to monitor inventory levels, warehouse conditions and enable real-time communication among warehouse staff.

Intelligent stores are more productive, which is one of their key advantages. Smart warehouses increase order fulfilment efficiency and precision by automating regular processes and evaluating real-time data to discover and solve inefficiencies. Labour expenses in intelligent warehouses can be decreased by automating labour-intensive tasks such as picking and packing. Smart warehouses outperform traditional warehouses regarding productivity, accuracy, and operating costs since they require fewer people. As a result, you'll see more of them in fields including manufacturing, retail, and logistics. The Internet of Things (IoT) has the potential to revolutionise the food processing industry by allowing for real-time monitoring and administration of multiple operations. The Internet of Things can be used in the food handling business, for example, by installing IoT intelligent sensors. Temperature, humidity, and microbial activity all impact food quality and safety. These devices are capable of monitoring and recording data on these factors. Using real-time data obtained by these sensors could reduce food waste and improve product quality. The Internet of Things (IoT) could track the movement of assets such as raw materials, equipment, and completed goods in the supply chain. This system could help food businesses manage inventory more effectively, reduce theft and loss, and optimise their supply chain [5] [6].

The Internet of Things (IoT) could help with predictive maintenance by tracking technology's health and predicting when it will need service. As a result, less care may be required, and productivity may increase. The Internet of Things (IoT) technology could help food processing companies use less energy, lowering operational costs and benefiting the environment. The Internet of Things (IoT) may aid in developing intelligent packaging systems capable of monitoring and preserving food quality and freshness. This has the potential to reduce food waste while also improving food safety. The Internet of Things (IoT) can increase the food processing industry's sustainability, productivity, and efficiency without compromising food safety or quality [7] [8].

III. MICROPYTHON AND ESP32

MicroPython is a version of Python 3 that is optimised for microcontrollers and other low-resource

devices. MicroPython is frequently used for hardware interface tasks, including reading sensor data and controlling actuators. Because of its powerful microcontroller with built-in Wi-Fi and Bluetooth connectivity and its vast array of sensors and interfaces, including SPI, I2C, and UART, the ESP32 is a good platform for developing Internet of Things applications in MicroPython. Its usability is enhanced by the fact that its development settings are user-friendly. The ESP32 is a low-cost, user-friendly instrument suitable for experts and beginners [9].

IV. FOOD DECAY MONITORING

A food deterioration tracking sensor can aid in identifying microorganisms that cause food to decay and monitoring the rate at which perishables spoil. The monitors could be used in a variety of ways across the food supply chain, from food production to distribution and storage. Carbon dioxide and ammonia, two gases created when items degrade, can be detected by gas detectors. Optical monitors use light to detect colour changes that may indicate spoilage or contamination of food. Electronic nose monitors can detect volatile molecules created by decaying bacteria to detect the early signs of food spoiling. The environment's temperature can affect how quickly food spoils during storage or transportation. Temperature sensors can thus be used to monitor and control the ambient temperature [10].

V. EDGE COMPUTING

Distributed computing, sometimes known as "edge computing," refers to putting computation and data storage closer to the data source. Data is managed and examined locally on edge devices or servers rather than transferred to a central data centre or cloud. Edge devices, which include smartphones, IoT devices, and sensors, are a large category of data-generating gadgets. Edge servers can be deployed at the network's edge or next-to-edge devices to be close to end users. Edge computing offers different benefits, including reduced latency, improved data safety and security, and lower bandwidth consumption. Because it processes data locally, edge computing could substantially cut the time required to learn something from data. This makes it excellent for real-time applications like self-driving cars, industrial automation, and smart cities. Edge computing has the potential to minimise bandwidth consumption and save money by reducing the amount of data transferred to centralised data hubs or the cloud. Edge computing, which processes data locally, can improve security and safety. This is because sensitive information is not necessary to be transferred via the Internet. Instead, it can be administered and inspected locally. Because it allows data to be processed and evaluated in near real-time at its source, edge computing represents a fundamental shift in the operation of distributed computing systems. This strategy enhances data privacy and security [11].

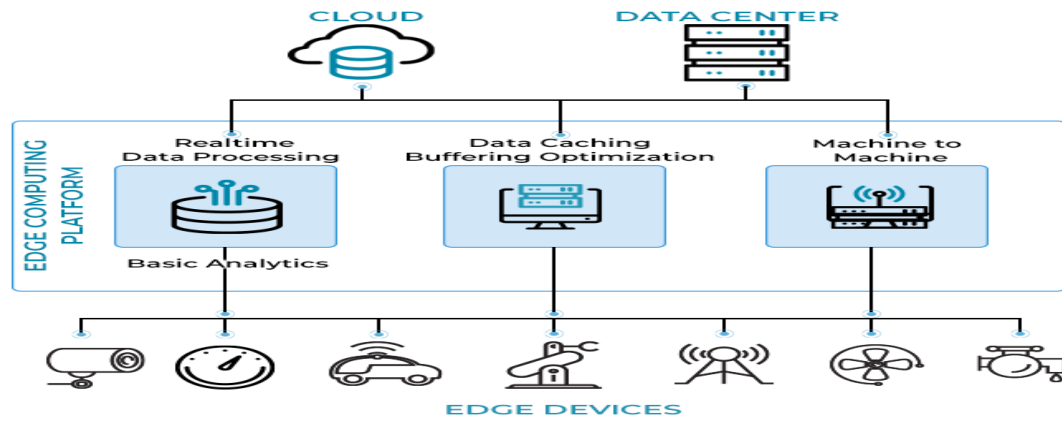


Figure 1. Edge Computing Architecture

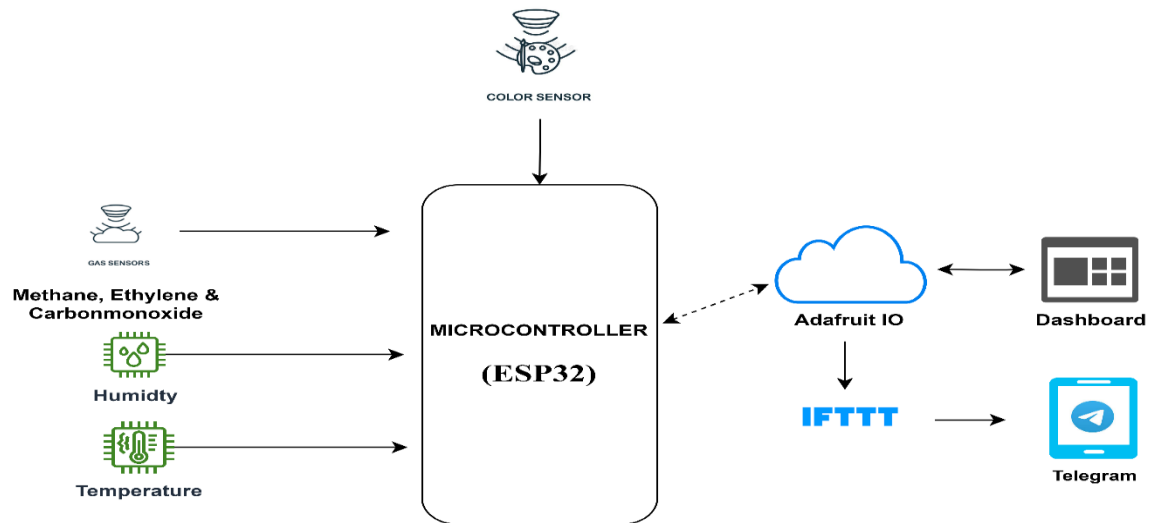


Figure 2. Architecture of Smart warehouse Shelves

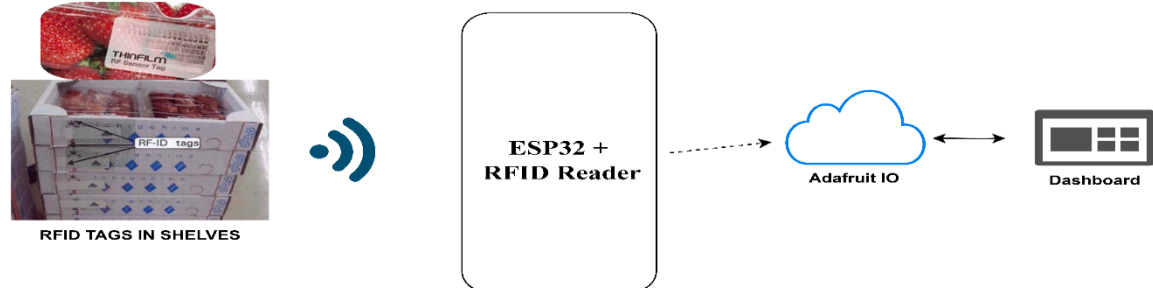


Figure 3. RFID structure for food items tracking

VI. ADVANTAGE OF EDGE COMPUTING OVER CLOUD COMPUTING

The exponential growth of data and its impact on network performance is one of the challenges with cloud computing. The real time data processing and analysis that is precise and fast. Edge computing outperforms cloud computing in terms of response time and real-time processing. The edge computing device may do calculations and store data locally due to its proximity to the data source, removing the need for data transmission. Data transmission speed, processing certainty in real time, and waiting periods are all improved by emphasising proximity to users and delivering intelligent, modern

services. Edge computing allows consumers to get quick responses to a wide range of services, particularly location-based services and video surveillance software. Low cost, low energy consumption and little network impact are the holy grails of system requirements. Edge computing eliminates the requirement for data processing to be sent to a remote cloud data centre. This minimises network traffic and smart device congestion [12] [13].

VII. EDGE COMPUTING ENABLED SMART WAREHOUSE SHELVES

Using a number of technologies and devices, edge computing can be utilised to inspect the status of food in a warehouse. This enables better storage conditions. The

current study presents a framework for assuring food safety and traceability, with a focus on vegetables. RFID technology allows you to track individual food items as well as full batches. Using an ESP32-based sensor node to monitor temperature, humidity, risk, and bacterial proliferation. Temperature and humidity are not the only variables that might have an impact on product quality. Light, air quality, and chemicals like ethylene gas generated by some facilities can all have an effect. Warehouse administrators can monitor and report on a variety of factors using ESP32-powered sensor nodes, and then take appropriate action. Figure 2 shows how clever warehouse stacking works structurally. It is now possible to monitor food storage and transportation using technology. With this system in place, food waste is reduced, inventory discrepancies are reduced, and recalls are easier to manage if food safety problems emerge. Warehouse managers can rapidly dispose of or shift goods if they monitor important characteristics such as expiration dates and signs of wear and tear [14] [15].

Figure 3 depicts the structure of the RFID tracking device for food items. RFID tags can be attached to raw materials, ingredients, and finished goods to help in tracking and inventory control. As a result, the supply chain is easier to manage, stockouts are reduced, and production processes are enhanced. RFID tags have the ability to store and retrieve product information such as batch number, manufacture date, expiration date, and quality. This not only guarantees that laws are followed, but it also makes it simple and quick to discover a recalled product for health or safety reasons. RFID technology allows for the administration of apparatus setup, automatic product sorting and packaging, and monitoring of the flow of raw materials and completed goods throughout the manufacturing process. If the IoT system detects an anomaly, such as a temperature change, the presence of toxic vapours, or the approaching of an expiration date, it may send out alerts and messages promptly. Alerts provided to warehouse staff mobile devices or a central dashboard for tracking make it easy for them to handle concerns immediately and preserve food quality [16] [17]. To assess the air quality, ethylene detectors are put on the shelves of intelligent warehouses. It is possible to build detectors that provide real-time information about the ethylene concentration in the air. Developers can construct their own applications and integrations using Telegram's Application Programming Interface (API). This API is compatible with IoT devices. If This Then That can serve as a link to your chat app [18] [19]. Before you can use Telegram, you must first establish an application and receive an API code. The token allows the middleware or programme to communicate with the bot, execute orders, and carry out other tasks on its behalf. Users on the popular web-based automation platform IFTTT can use applets to create simple conditional statements that connect and orchestrate services like Telegram. You may use IFTTT to connect Telegram to a number of different programmes, web services, and tools in order to construct your own automations and workflows. Complex automation routines and integrations with Telegram and other services can be created using a large number of applets and a mix of

triggers and actions [20] [21] [22]. Figure 4 shows the working of the smart workhouse system with edge computing capabilities.

VIII. USER-ORIENTED INTERFACE

It has been developed with a user-friendly layout to ensure that it looks and operates properly. The primary interface of the application shows how hazard levels in the storage facility are monitored [23] [24]. An extensible tab starts and stops monitoring, stores measurement data, and sends it to a device that can read it. The temperature, humidity, and quantity of various gases are measured and displayed using a pie chart. In addition to tracking, local data may be shared with others via Telegram groups and subsequently uploaded to Adafruit IO's cloud service. People can analyzed and evaluate data and its consequences on a specific platform using a variety of electronic tools, such as desktop computers and mobile applications [25] [26].

IX. EXPERIMENT DESIGN AND DEPLOYMENT

The method will be thoroughly examined and tested to guarantee its effectiveness and robustness. The method will be thoroughly examined and tested to guarantee its effectiveness and robustness. To thoroughly analyse the sensor node, use the ESP32's real-time data and functionality. Examine the system's load and functionality. An in-depth examination of RFID and sensor nodes.

Figure 7 displays the project's layout. For usage in intelligent warehouse shelves, a Micropython-programmed ESP32-based hardware system capable of perception and communication was built.

The prototype could be used to improve human-machine communication by being applied to a variety of sensor terminals. Adafruit IO hosts data analyses that are completer and more usable over longer time periods. Furthermore, the Python database is critical for the system's durability and utility. The intelligent warehouse's dependence on the If This Then That (IFTTT) protocol, which is supplemented by data visualisation and sharing features, enables Adafruit IO data communication. A website provides users with access to the graphical administration and information exchange interface. Figure 6 depicts a display that tracks the location of each tier in the structure.

The 15 foods mentioned above were chosen to ensure that the results were accurate and reliable. If different food sets were investigated, one for each type of monitor in the warehouse node. The calculated average number was the final solution. Use the If This Then That (IFTTT) Telegram plugin to configure your conversations to trigger specific actions. Instead of relying simply on email for external communication and automation, RESTful and IFTTT support allow you to examine a broader range of possibilities. Furthermore, the speed and dependability of these technologies are incredibly high. After installation, the data acquisition procedure is completely automated, removing the need to submit data files manually. The IFTTT framework's real-time application programming interfaces (APIs) shorten an applet's response time whenever a sensor indicates a change. Monitors that use a

real-time API must report any change, even if it is unrelated to the triggering event. As a result, communications are transmitted when they are not necessary. Figure 7 shows the warning letter sent to the appropriate recipients via Telegram channels and groups. Applets that connect Telegram to other services can be constructed using Telegram and the interface offered by If This Then That. Following that, we'll show you how to connect Telegram and IFTTT. Alerts require your immediate attention. Some

apps allow you to receive notifications from several sources on your Telegram account. Upload individual Telegram messages or cloud-based multimedia such as images, movies, or files using applets. Applets can be set up to do this on their right away. Users can interact with the bot and initiate activities in other IFTTT-connected services by creating interactive workflows. After configuring the applets, users will get notifications or take actions via Telegram based on their preferences.

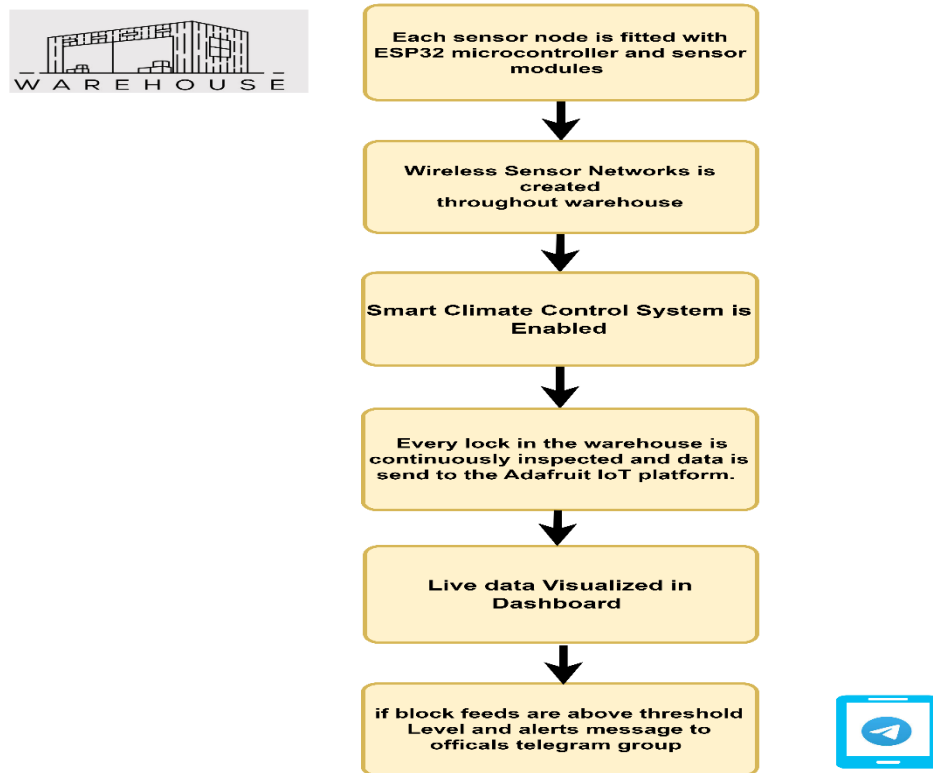


Figure 4. Attendance Monitoring based on the Radio Frequency Identification and IoT

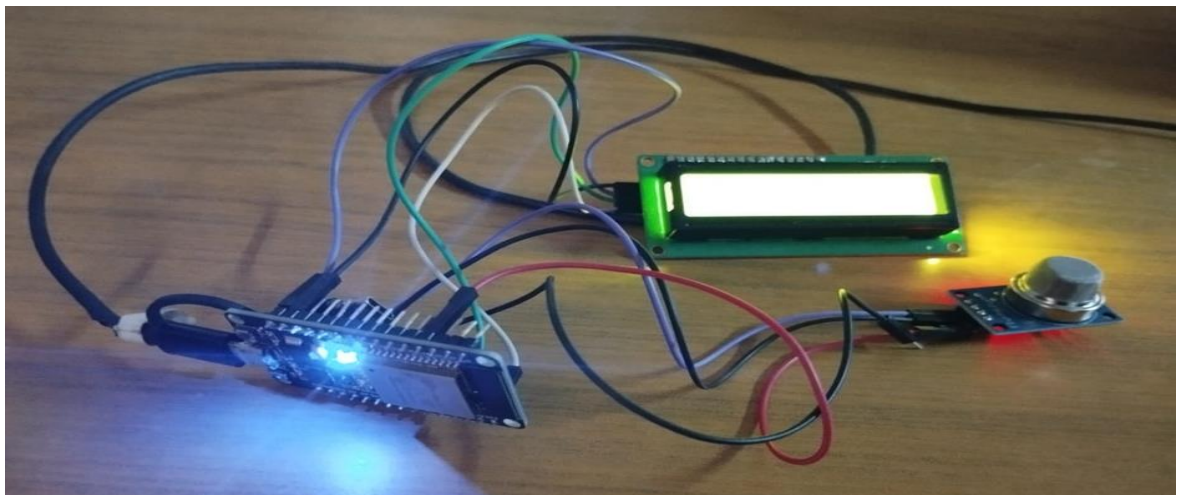


Figure 5. Experimental Setup for Measuring Toxic Level of Warehouse Shelves

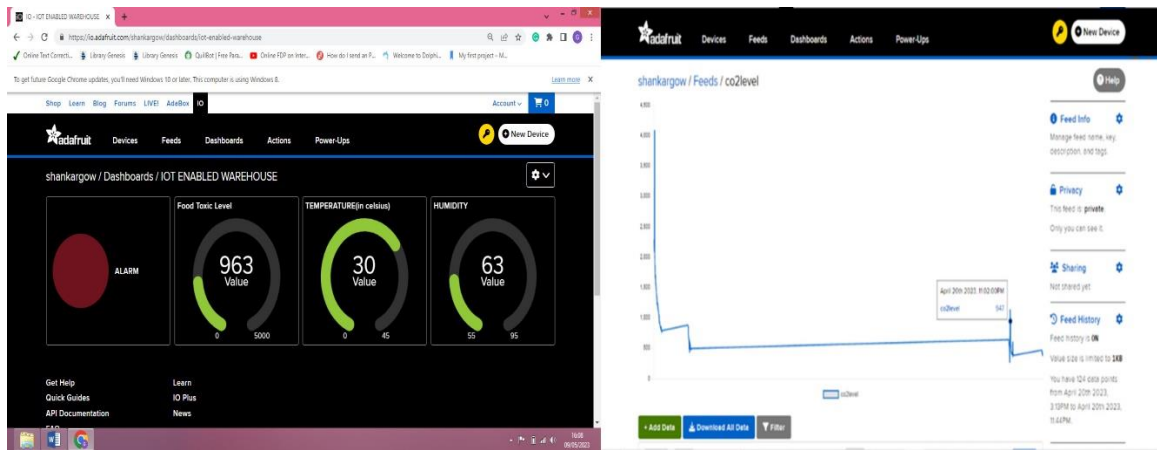


Figure 6. Dashboard Monitoring of each shelves of Warehouse

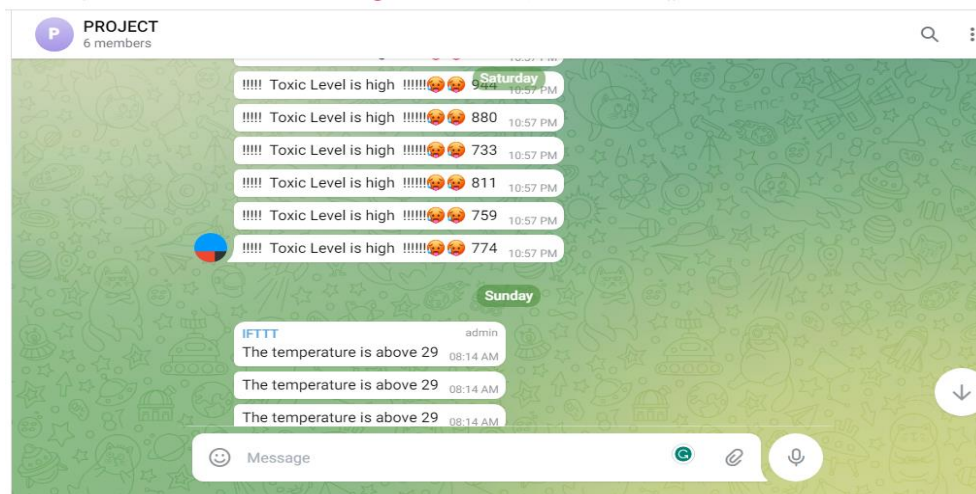


Figure 7. Alert notification in Telegram channel/groups to officials

X. CONCLUSION

Edge computing refers to locating computers, data storage, and intelligent Internet services to the network's outskirts. This strategy meets the many data needs of today's sectors, making it easier for them to go digital. The warehouse system can constantly monitor fragile objects on the racks thanks to the Internet of Things (IoT). As part of this monitoring, we look for contaminants in the food and monitor elements like temperature and humidity. This research aims to integrate RFID technology with edge computing to create intelligent warehouse management systems for the agriculture and food processing industries. Real-time monitoring of warehouse shelves is possible, and users can receive notifications via telegram or smartphone. This research aims to create and deploy food-tracking systems using wireless sensor technology, database technology, and other relevant technologies. Fruits are utilized in the study to demonstrate how these techniques can be implemented. The prototype developed for this project has many development opportunities.

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